



Hinged Flood Mitigating Storm Drain

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Problem/Problem Statement

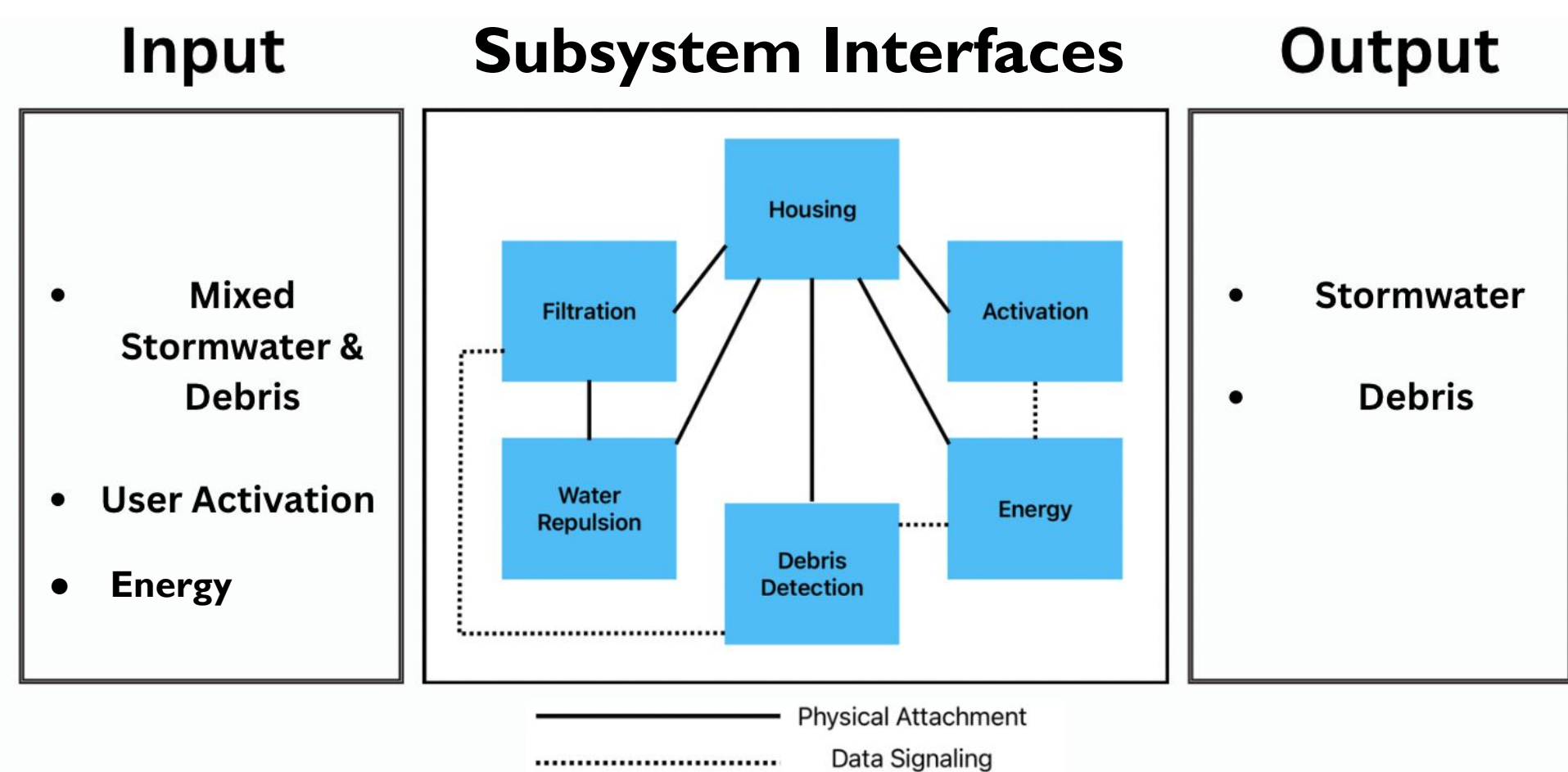
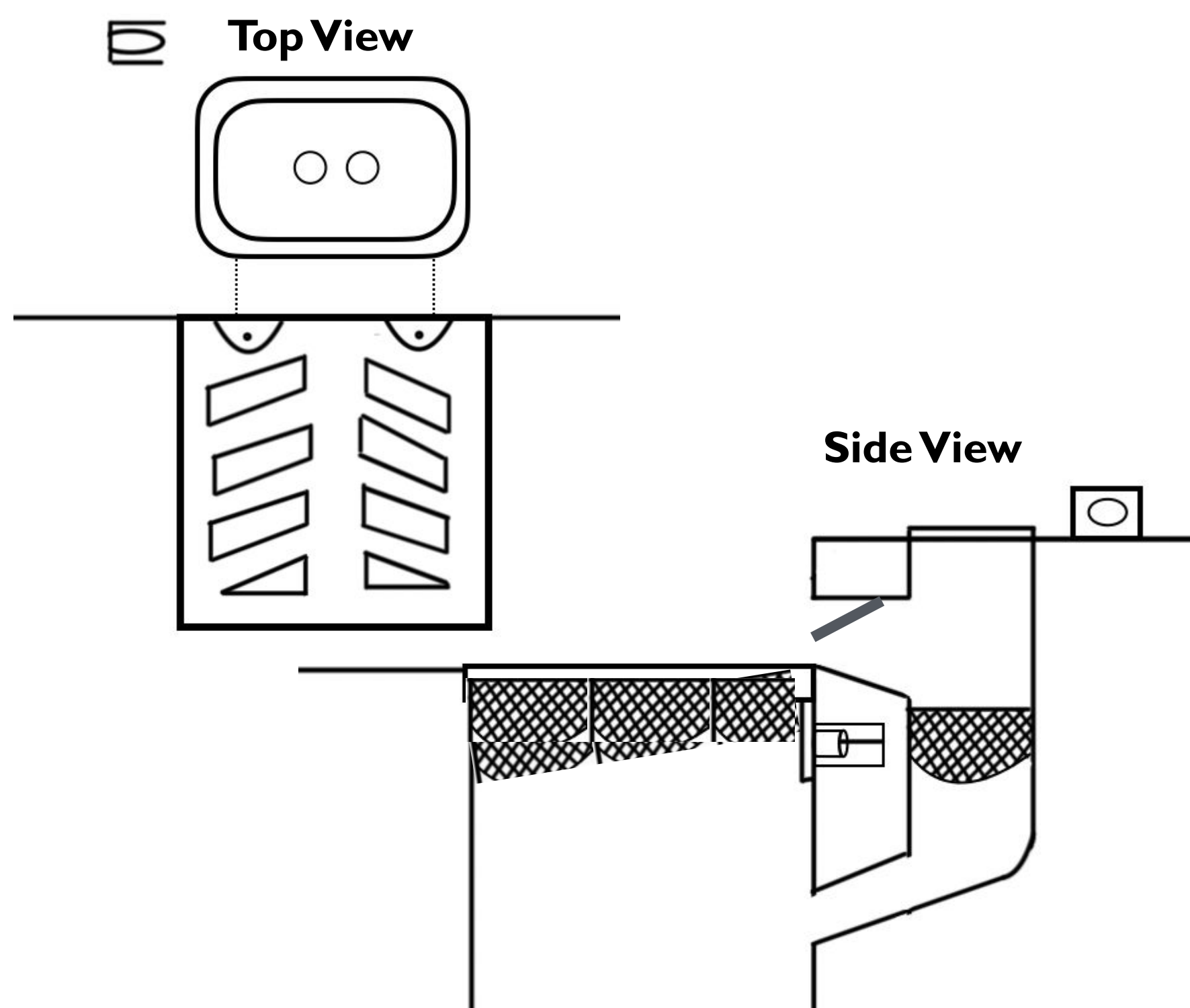
The last few years have had a high amount of floods causing devastation and destruction. We have been tasked to create a solution which address some challenge brought by flooding. Additionally, it is crucial that we are socially considerate and follow proper ethics with our design. Our problem statement is as follows, “Many aspects of San Francisco's current flood mitigation systems are increasingly less equipped to manage the growing flood risks due to climate change. Revisiting aging and underdeveloped infrastructure is a necessity to protect the cultural and human capital of particularly at-risk communities, given their particular geographic and socioeconomic constraints.”

Design Summary

- Our final design is an advanced hinged storm drain that allows for fast, safe and efficient removal of accumulated debris during flood events.
- To do this our storm drain has a tilting grate at the mouth, foot-pedal activation on top of the street, and a debris collection tank below the street. Once activated the grate is lifted by heavy-duty hinges, allowing the debris to enter the grate opening and slide into the collection tank. While this is happening water continues down the drain unobstructed.
- This solution reduces clogging, increased stormwater flow, and eases maintenance for public workers.

Risk Management

	Minor	Significant	Major	Severe
Most likely	Basket filling up	Lingering raindrops keeping the gate open	The AI gives incorrect basket capacity %	Large debris blocking the gate from closing post flood
Pretty likely	Random water randomly triggering the sensor	An untrained professional would try to “take out the trash”	Uneducated stakeholder improper usage	Electrical Hydraulics malfunctioning
Possible	A too big object blocks the gate	The cover gets stuck	Sensor malfunctioning	Failure to open storm gate
Not very likely	Insects disrupting drain function	AI giving incorrect signals	Gate not opening	Power cutting off



Solution Validation

Test Results

- Tested the basket with rocks, leaves, and debris to simulate real storm conditions.
- Major of debris was captured by the grate and mesh while water continued draining through
- The basket held the weight of rocks without deforming.
- Leaves/light debris stayed contained and did not pass through.
- Basket was easy to remove and empty, confirming maintenance efficiency and easy to use for stakeholders..

Key Subsystems

- **Housing**
 - Physically connects and supports all subsystems.
 - Anchors grate, hinges, pedal, and collection tank
- **Activation**
 - Allows user to activate storm drain gate using foot pedal
 - Hinges enable grate to rotate into position
- **Energy**
 - Supplies power to Activation & debris-detection sonar
- **Filtration**
 - Separates solids from flood water using a galvanized grate and mesh.
 - Collection basket is filtered to allow excess water to escape
- **Water Repulsion**
 - Directs unobstructed, filtered water into pre-existing sewer system
- **Debris Detection**
 - Signals when the basket is full using sonar sensor

References

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